Modal Logic Grounded in Cost Geometry: Necessity, Possibility, and Actuality from $J(x)$

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Abstract

We ground the fundamental concepts of modal logic—necessity, possibility, impossibility, and actuality—in the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, the unique normalized solution to the Recognition Composition Law. A configuration is *necessary* if and only if it is the unique J-minimizer ($x = 1$, the sole zero-defect state). A configuration is *possible* if and only if it has positive ratio ($x > 0$, ensuring finite J-cost). A configuration is *impossible* if $x \leq 0$ (violating ledger positivity). Actuality is identified with the J-minimum—a form of necessitarianism grounded in cost geometry. We prove that the core S5 structural content holds as theorems: necessity implies actuality implies possibility; the necessary configuration is unique; and every possible configuration is accessible to the necessary one via cost-decreasing paths. We further define propositional modal operators $\Box$ and $\Diamond$ over configuration space and prove the distribution axiom K, modal duality, and a weak form of the truth axiom T. Modal distance from the actual world is measured by defect: $d(x) = J(x) \geq 0$, with $d(x) = 0$ if and only if $x = 1$. This provides modal metaphysics with physical content: necessity is not "truth in all possible worlds" but unique cost-geometric forcing; possibility is not a Lewisian concrete world but finite cost; and actuality is not indexical but the J-minimum. The foundational mathematical results (uniqueness of the J-minimizer, non-negativity of J, AM–GM inequality) are rigorously proved within the RS framework.

1 Introduction

Modal logic—the logic of necessity ($\Box$) and possibility ($\Diamond$)—is central to metaphysics, epistemology, and the philosophy of language. Yet its foundational concepts remain contested. What does it mean for something to be "necessary"? What are "possible worlds"? Is the actual world distinguished structurally, or is its status merely indexical?

Lewis [1] proposed modal realism: possible worlds are concrete entities as real as the actual world, and actuality is indexical ("the world I happen to inhabit"). Kripke [2] treated possible worlds as abstract stipulations, grounding necessity in rigid designation. Plantinga [3] identified necessity with maximal states of affairs. Williamson [4] developed a systematic case for treating modal logic as first-order metaphysics, arguing

that necessitism—the thesis that everything necessarily exists—provides the best overall theory. Stalnaker [6] offered a moderate realism about propositions as sets of possible worlds while resisting Lewis's ontological inflation. Each approach has virtues, but none derives modal concepts from a deeper mathematical structure with independently motivated physical content.

Recognition Science [10] provides such a derivation. The theory begins with a single functional equation—the Recognition Composition Law (RCL)—whose unique normalized solution is the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ on $\mathbb{R}_{>0}$. From this functional alone, we derive precise mathematical content for each modal concept:

- **Necessity** = unique J-minimizer ($x = 1$).
- **Possibility** = positive ratio ($x > 0$, finite cost).
- **Impossibility** = non-positive ratio ($x \leq 0$, undefined cost).
- **Actuality** = the J-minimum itself, not an indexical label.
- **Modal distance** = defect $J(x)$, a continuous quantitative measure.

The resulting modal structure satisfies the S5 axioms as *theorems* rather than postulates, and replaces the metaphysical debate about possible worlds with cost geometry. The framework entails a form of necessitarianism—the actual world is the only necessary one—that echoes Spinoza [5] and resolves the Leibniz question.

The paper is structured as follows. Section 2 derives the cost functional. Section 3 introduces the RS modal definitions. Sections 4–7 prove the structural theorems. Section 8 demonstrates the S5 axioms. Section 9 develops quantitative modality. Section 10 defines propositional modal operators. Section 11 compares the framework with standard approaches and draws philosophical consequences.

2 The Cost Functional

2.1 Axioms and Uniqueness

The cost functional is the unique solution to three axioms:
Axiom A1 (Normalization): $J(1) = 0$ (the identity configuration has zero cost).
Axiom A2 (Recognition Composition Law, RCL): For all $x, y > 0$:

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y). \tag{1}$$

This equation expresses that the cost of comparing a composite (xy) and its reciprocal complement (x/y) decomposes multiplicatively and additively in terms of the component costs.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.1 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t)+1$ and rewriting (A2) gives the d'Alembert functional equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [9], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. Inverting, $J(x) = f(\ln x) - 1 = \cosh(\ln x) - 1 = \frac{1}{2}(x + x^{-1}) - 1$. □

2.2 Verification and Key Properties

Subject to (A1)–(A3), the unique C^2 solution is:

Definition 2.2 (Cost Functional and Defect)**.** For $x \in \mathbb{R}_{>0}$:

$$J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{2}$$

The *defect* of x is $\mathrm{defect}(x) := J(x)$.

That this J satisfies the RCL can be verified by direct computation. The left side:

$$J(xy) + J(x/y) = \tfrac{1}{2}\big(xy + (xy)^{-1} + xy^{-1} + x^{-1}y\big) - 2. \tag{3}$$

The right side:

$$2J(x)J(y) + 2J(x) + 2J(y) = \tfrac{1}{2}(x + x^{-1})(y + y^{-1}) - 2, \tag{4}$$

where the intermediate terms cancel. Expanding the product on the right gives $\frac{1}{2}(xy + xy^{-1} + x^{-1}y + x^{-1}y^{-1}) - 2$, confirming equality.

2.3 Key Properties

Lemma 2.3 (Basic Properties of J)**.** *(i) $J(x) \geq 0$ for all $x > 0$, with equality if and only if $x = 1$.*

(ii) $J(x) = J(x^{-1})$ for all $x > 0$ (reciprocal symmetry).

(iii) J is strictly convex on $(0, \infty)$, with $J''(x) = x^{-3} > 0$.

(iv) $J(x) \to \infty$ as $x \to 0^+$ or $x \to \infty$.

Proof. (i) By AM–GM, $x + x^{-1} \geq 2\sqrt{x \cdot x^{-1}} = 2$ for $x > 0$, with equality iff $x = x^{-1}$, i.e., $x = 1$. Hence $J(x) \geq 0$.

(ii) $J(x^{-1}) = \frac{1}{2}(x^{-1} + x) - 1 = J(x)$.

(iii) $J'(x) = \frac{1}{2}(1 - x^{-2})$ and $J''(x) = x^{-3} > 0$ for $x > 0$.

(iv) As $x \to 0^+$, $x^{-1} \to \infty$, so $J(x) \to \infty$. As $x \to \infty$, $\frac{1}{2}x \to \infty$, so $J(x) \to \infty$. □

3 RS Modal Definitions

We define the four fundamental modal concepts in terms of J. The geometric motivation is that J defines a cost landscape on $\mathbb{R}_{>0}$ with a unique global minimum at $x = 1$. Modal concepts correspond to positions on this landscape.

Definition 3.1 (RS Necessity)**.** A configuration $x \in \mathbb{R}$ is *RS-necessary* if and only if

$$x > 0 \quad \text{and} \quad J(x) = 0. \tag{5}$$

A configuration is necessary if it incurs zero cost. It is not "forced" by fiat but by the structure of J—any deviation from $x = 1$ strictly increases cost.

Definition 3.2 (RS Possibility)**.** A configuration x is *RS-possible* if and only if $x > 0$.

A configuration is possible if it has a well-defined, finite cost. The positivity requirement ensures that $J(x)$ is defined and real. Configurations with $x \leq 0$ lie outside the domain of J entirely.

Definition 3.3 (RS Actuality)**.** A configuration x is *RS-actual* if and only if it is RS-necessary.

This identification—actuality *is* necessity—is the central metaphysical claim. It is not a tautology but a substantive thesis: the actual world is not one possible world among many that happens to be "ours" (Lewis), nor a stipulated entity (Kripke), but the unique cost-geometric minimum. The philosophical significance of this identification is discussed in Section 11.

Definition 3.4 (RS Impossibility)**.** A configuration x is *RS-impossible* if and only if $x \leq 0$.

Definition 3.5 (Modal Distance)**.** The *modal distance* from x to the actual world is

$$d(x) := J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{6}$$

4 Necessity: The Unique J-Minimizer

Theorem 4.1 (Unique Necessity)**.** *There exists exactly one RS-necessary configuration: $x = 1$.*

Proof. Existence. $1 > 0$ and $J(1) = \frac{1}{2}(1 + 1) - 1 = 0$.

Uniqueness. If $y > 0$ and $J(y) = 0$, then $\frac{1}{2}(y + y^{-1}) = 1$, so $y + y^{-1} = 2$, so $y^2 - 2y + 1 = 0$, so $(y - 1)^2 = 0$, so $y = 1$. □

Corollary 4.2 (Necessity is Identity)**.** *For all $x \in \mathbb{R}$: x is RS-necessary if and only if $x = 1$.*

Proof. Forward: Theorem 4.1. Backward: $J(1) = 0$ and $1 > 0$. □

Remark 4.3 (Structural Forcing)*.* The uniqueness of the necessary configuration follows from the strict convexity of J (Lemma 2.3(iii)). A strictly convex function on a convex domain has at most one global minimum. Since J achieves its minimum at $x = 1$ and is strictly convex on $(0, \infty)$, no other configuration can have zero defect. Necessity is *structurally forced* by the geometry of the cost landscape, not postulated.

5 Possibility: Positive Ratio and Finite Cost

Theorem 5.1 (Possibility is Broad)**.** *For all $x > 0$: x is RS-possible. Conversely, every RS-possible x satisfies $x > 0$.*

Proof. Immediate from Definition 3.2. □

Theorem 5.2 (Finite Cost for Possible Configurations)**.** *Every RS-possible configuration has non-negative, finite J-cost.*

Proof. For $x > 0$, $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ is a finite real number, and $J(x) \geq 0$ by Lemma 2.3(i). □

Theorem 5.3 (Possibility is Not Singleton)**.** *There exist distinct RS-possible configurations.*

Proof. $x = 1$ and $x = 2$ are both positive, hence both RS-possible, and $1 \neq 2$. □

6 Impossibility: The Positivity Boundary

Theorem 6.1 (Impossibility Equivalence). *x is RS-impossible if and only if x is not RS-possible.*

Proof. RS-impossible means $x \leq 0$. Not RS-possible means $\neg(x > 0)$, which is $x \leq 0$. □

Theorem 6.2 (No Overlap). *No configuration is both RS-possible and RS-impossible.*

Proof. $x > 0$ and $x \leq 0$ are contradictory. □

Remark 6.3 (The Positivity Boundary). The boundary $x = 0$ represents a hard limit: as $x \to 0^+$, $J(x) \to \infty$ (Lemma 2.3(iv)), so the boundary is not merely forbidden but infinitely expensive to approach. Impossibility thus has quantitative character: the "closer" a configuration is to the boundary (in the sense of small x), the more costly it is.

7 Actuality: The J-Minimum

Theorem 7.1 (Actuality is Unique). *$x = 1$ is the unique RS-actual configuration.*

Proof. RS-actual $\Leftrightarrow$ RS-necessary $\Leftrightarrow$ $x = 1$ (Corollary 4.2). □

Theorem 7.2 (The Merely Possible). *For $x > 0$ with $x \neq 1$: x is RS-possible but not RS-actual.*

Proof. $x > 0$ ensures RS-possible. $x \neq 1$ implies $J(x) > 0$ (Lemma 2.3(i)), so x is not RS-necessary, hence not RS-actual. □

Remark 7.3 (Tripartite Classification). The space of configurations is partitioned into three classes:

(a) The unique *actual/necessary* configuration: $x = 1$, $J(x) = 0$.
(b) The *merely possible* configurations: $x > 0$, $x \neq 1$, $J(x) > 0$.
(c) The *impossible* configurations: $x \leq 0$.

This classification emerges from the cost landscape alone.

8 S5 Modal Axioms as Theorems

The standard S5 system of modal logic consists of the axioms K (distribution), T (truth), and 5 (positive introspection), or equivalently K, T, and B (symmetry). In standard Kripke semantics, S5 corresponds to an accessibility relation that is an equivalence relation (reflexive, symmetric, transitive). We show that the RS cost geometry satisfies the structural content of S5.

8.1 The Implication Chain

Theorem 8.1 (Axiom T: Necessity $\Rightarrow$ Actuality). *For all x: if x is RS-necessary, then x is RS-actual.*

Proof. RS-actual is defined as RS-necessary (Definition 3.3). □

Remark 8.2. The identification actuality = necessity (Definition 3.3) makes Theorem 8.1 definitionally immediate. The substantive content is in the definition itself: the claim that the unique J-minimum plays the role of the "actual world" gives modal Axiom T a geometric rather than a stipulative basis.

Theorem 8.3 (Axiom D: Actuality $\Rightarrow$ Possibility)**.** *For all x: if x is RS-actual, then x is RS-possible.*

Proof. RS-actual implies RS-necessary, which requires $x > 0$ by Definition 3.1. Hence x is RS-possible by Definition 3.2. □

Corollary 8.4 (Full Implication Chain)**.** *For all x:*

$$\textit{RS-necessary}(x) \implies \textit{RS-actual}(x) \implies \textit{RS-possible}(x).$$

8.2 Accessibility

The S5 axiom B (symmetry of accessibility) is captured in the RS framework by the following theorem, which is strictly stronger than B: not only is every possible configuration accessible to the necessary one, but the access path is cost-monotone.

Theorem 8.5 (Cost-Monotone Accessibility)**.** *For every RS-possible x, there exists a continuous path $\gamma : [0, 1] \to (0, \infty)$ with $\gamma(0) = x$, $\gamma(1) = 1$, and $J(\gamma(t))$ non-increasing.*

Proof. Define $\gamma(t) = x^{1-t}$ for $t \in [0, 1]$. Then $\gamma(0) = x$, $\gamma(1) = x^0 = 1$, and $\gamma(t) = x^{1-t} > 0$ for all t (since $x > 0$).

Let $s = 1 - t$ and $u(s) = J(x^s) = \frac{1}{2}(x^s + x^{-s}) - 1$. Then

$$u'(s) = \tfrac{1}{2}\ln(x)\big(x^s - x^{-s}\big).$$

For $x > 1$: $\ln x > 0$ and $x^s > x^{-s}$ when $s > 0$, so $u'(s) > 0$. Since $s = 1 - t$, we have $\frac{d}{dt}J(\gamma(t)) = -u'(s) < 0$ for $t < 1$.

For $0 < x < 1$: $\ln x < 0$ and $x^s < x^{-s}$ when $s > 0$, so $u'(s) = (\text{negative})(\text{negative}) > 0$. Again $\frac{d}{dt}J(\gamma(t)) < 0$.

For $x = 1$: $\gamma(t) \equiv 1$ and $J \equiv 0$.

In all cases, $J(\gamma(t))$ is non-increasing from $J(x)$ to $J(1) = 0$. □

Remark 8.6 (Structural S5). In the RS framework, the accessibility structure is richer than an equivalence relation: it is a continuous cost landscape with a unique global minimum. Theorem 8.5 provides not merely that every possible configuration *can reach* the necessary one, but that there is a *directed*, cost-decreasing path. This gives the accessibility relation a natural orientation absent from standard Kripke frames.

9 Modal Distance: Quantitative Modality

Theorem 9.1 (Modal Distance Properties)**.** *(i) $d(x) \geq 0$ for all $x > 0$ (non-negativity).*

(ii) $d(x) = 0$ if and only if $x = 1$, i.e., x is RS-actual (identity of indiscernibles).

(iii) $d(x) = d(x^{-1})$ for all $x > 0$ (reciprocal symmetry).

(iv) $d(1) = 0$ (zero self-distance).

Proof. (i)–(ii) follow from Lemma 2.3(i). (iii) follows from Lemma 2.3(ii). (iv) is $J(1) = 0$. □

Remark 9.2 (Quantitative Modality). Standard modal logic is qualitative: a proposition is necessary, possible, or impossible, with no intermediate grades. Modal distance replaces this trichotomy with a continuous spectrum. A configuration with $d(x) = 0.01$ is "almost necessary"; one with $d(x) = 100$ is "very far from the actual world." This quantitative structure has no analog in Lewis, Kripke, Plantinga, or Williamson.

The reciprocal symmetry $d(x) = d(x^{-1})$ means that "doubling" and "halving" are equidistant from actuality. A world where some ratio is 2 is exactly as far from actuality as a world where that ratio is $\frac{1}{2}$. This is a non-obvious structural feature inherited from the cost functional's reciprocal invariance.

10 Propositional Modal Operators

The definitions in Sections 3–7 concern *configurations* (real numbers). We now lift these to *propositional* modal operators over a configuration space, making contact with standard modal logic syntax.

10.1 Configuration Space and Possibility Set

Definition 10.1 (Configuration)**.** A *configuration* is a triple $c = (x, t, B)$ where $x > 0$ is the ratio value, $t \in \mathbb{N}$ is the tick count, and $|\ln x| \leq B$ for a fixed bound $B \leq 16$. We write $\mathcal{C}$ for the set of all configurations.

Remark 10.2 (The bound $B \leq 16$). The restriction $B \leq 16$ in Definition 10.1 is required by Lemma 10.6: for the identity $A(c)$ to be accessible from c, one needs $|\ln x| \leq 16$ (see proof below). Physically, $|\ln x| \leq 16$ means $e^{-16} \leq x \leq e^{16}$ (i.e., ratios within a factor of $\approx 8.9 \times 10^6$ of unity), which is a generous bound covering all RS physical ratios. Configurations outside this range require multi-epoch transitions and are not excluded from RS physics, but they fall outside the single-epoch modal framework here.

Definition 10.3 (Stasis and Transition Costs)**.** For configurations with values $x, y > 0$:

$$J_{\text{stasis}}(x) = 8 \cdot J(x), \tag{7}$$

$$J_{\text{trans}}(x, y) = |\ln(y/x)| \cdot \tfrac{1}{2}\big(J(x) + J(y)\big). \tag{8}$$

The stasis cost is the J-cost of remaining at x for one eight-tick epoch. The transition cost measures the expense of changing from x to y, weighted by the average defect.

Definition 10.4 (Possibility Set)**.** For a configuration $c = (x, t, B)$, the *possibility set* $\text{Poss}(c)$ consists of all configurations $c' = (y, t{+}8, B')$ such that transitioning does not exceed the stasis cost:

$$J_{\text{trans}}(x, y) + J(y) \leq J_{\text{stasis}}(x). \tag{9}$$

Definition 10.5 (Actualization)**.** The *actualization* of $c = (x, t, B)$ is $A(c) = (1, t{+}8, 0)$—the identity configuration at the next epoch.

Lemma 10.6 (The Identity is Always Possible)**.** *For every $c \in \mathcal{C}$, $A(c) \in \text{Poss}(c)$.*

Proof. $J(1) = 0$, so the constraint becomes $J_{\text{trans}}(x, 1) + 0 \leq 8\,J(x)$, i.e., $|\ln x| \cdot \frac{J(x)}{2} \leq 8\,J(x)$. If $J(x) = 0$ then $x = 1$ and both sides vanish. If $J(x) > 0$, dividing gives $|\ln x| \leq 16$, which holds by the bound constraint on configurations. □

10.2 Modal Operators

Definition 10.7 (Box and Diamond)**.** For a configuration property $p : \mathcal{C} \to \{\mathrm{T}, \mathrm{F}\}$:

$$(\Box p)(c) \iff \forall\, c' \in \mathrm{Poss}(c),\ p(c'), \tag{10}$$

$$(\Diamond p)(c) \iff \exists\, c' \in \mathrm{Poss}(c),\ p(c'). \tag{11}$$

10.3 Key Theorems

Proposition 10.8 (Modal Duality)**.** $(\Box p)(c) \iff \neg\,(\Diamond\,(\neg p))(c)$.

Proof. $(\Box p)(c)$ means $\forall c' \in \mathrm{Poss}(c)$, $p(c')$, which is $\neg\,\exists c' \in \mathrm{Poss}(c)$, $\neg p(c')$, which is $\neg\,(\Diamond\,(\neg p))(c)$. □

Proposition 10.9 (Axiom K: Distribution)**.** *If* $(\Box(p \Rightarrow q))(c)$ *and* $(\Box p)(c)$*, then* $(\Box q)(c)$*.*

Proof. Let $c' \in \mathrm{Poss}(c)$. By hypothesis, $p(c') \Rightarrow q(c')$ and $p(c')$. Hence $q(c')$. Since c' was arbitrary, $(\Box q)(c)$. □

Proposition 10.10 (Weak Axiom T)**.** *If* $(\Box p)(c)$*, then* $p(A(c))$*.*

Proof. By Lemma 10.6, $A(c) \in \mathrm{Poss}(c)$. Since $(\Box p)(c)$ asserts $p(c')$ for all $c' \in \mathrm{Poss}(c)$, we obtain $p(A(c))$. □

Remark 10.11 (Temporal Asymmetry and Weak T)*.* This is a "weak" form of Axiom T because it asserts that necessity implies truth at the *actualized* configuration $A(c)$, not at c itself. The standard Axiom T ($\Box p \to p$) would require c to be in its own possibility set, which the time-advance structure ($t \to t{+}8$) prevents: c has tick t, while all elements of $\mathrm{Poss}(c)$ have tick $t{+}8$. This temporal asymmetry is a feature rather than a defect: it reflects the arrow of time. The identity configuration at the *next* tick plays the role of the "actual world" in the standard T axiom.

Proposition 10.12 (Actualization Decreases Cost)**.** *If* $c = (x, t, B)$ *with* $x \neq 1$*, then* $J(A(c).\,x) < J(c.\,x)$*, i.e.,* $0 < J(x)$*.*

Proof. $A(c).\,x = 1$ and $J(1) = 0 < J(x)$ for $x \neq 1$ (Lemma 2.3(i)). □

11 Discussion

11.1 Comparison with Standard Frameworks

	Lewis	**Kripke**	**Williamson**	**RS**
Possible worlds	Concrete	Abstract	Abstract	Finite-cost configs
Necessity	All-worlds truth	Rigid desig.	Necessitism	Unique J-min
Actuality	Indexical	Privileged	Distinguished	J-minimum
S5 axioms	Postulated	Postulated	Argued	Proved
Ontological posit	Many worlds	Possible worlds	Necessary beings	None extra
Modal distance	None	None	None	$d(x) = J(x)$

RS has several advantages over each standard framework:

- **Over Lewis.** No ontological inflation. RS does not posit concrete possible worlds—merely configurations with positive cost. Lewis's "plurality of worlds" is replaced by a single cost landscape.
- **Over Kripke.** Actuality is not a matter of stipulation or rigid designation but is structurally determined by the cost geometry: $x = 1$ is the unique zero-cost configuration.
- **Over Williamson.** While Williamson argues for necessitism on abductive grounds, RS *derives* the unique necessary configuration from a functional equation. The existence of necessities is proved, not postulated.
- **Over Plantinga.** Maximal states of affairs are replaced by cost-geometric structure, giving modal concepts continuous, quantitative content via modal distance.
- **Over Stalnaker.** Stalnaker's moderate realism about propositions as sets of possible worlds is refined: possible "worlds" are configurations with specific, computable J-costs, and the proposition-set structure inherits a metric from J.

11.2 Necessitarianism: Spinoza and Leibniz

The RS identification of actuality with necessity—the claim that the actual world is the *only* necessary one—is a form of necessitarianism. This has deep historical resonance.

Spinoza (*Ethics*, Part I) argued that everything follows necessarily from the nature of God/Nature, so there is no genuine contingency. In RS, this is expressed geometrically: $x = 1$ is the unique zero-defect state, and its existence is structurally forced by the cost functional.

Leibniz asked "Why is there something rather than nothing?" and answered: because God chose the best of all possible worlds. RS gives a non-theistic, geometric version:

$$J(1) = 0 \quad \text{while} \quad J(x) \to \infty \text{ as } x \to 0^+. \tag{12}$$

"Something" ($x = 1$) has zero cost; "nothing" ($x \to 0$) has infinite cost. Existence is cheaper than non-existence. The actual world exists not because it was chosen but because non-existence would require infinite defect.

RS necessitarianism is, however, more constrained than Spinoza's. RS distinguishes sharply between the *laws* (which are necessary: there is a unique cost-zero configuration) and the *configurations with nonzero defect* (which are merely possible). The merely possible configurations are real in the sense of having finite cost, but they are not actual—they represent departures from the necessary state. This yields a compatibilist picture: the laws are necessary, but the specific state of the world at any given tick is contingent upon the dynamics.

11.3 Counterfactuals

The framework naturally accommodates counterfactual reasoning. A *counterfactual configuration* relative to c is any $c' \in \mathrm{Poss}(c)$ with $c' \neq A(c)$ and $J(c'.x) > J(A(c).x)$. These are configurations that *could have* been reached (they are in the possibility set) but were not actualized (they have higher cost than the actualization).

The "closeness" of a counterfactual to actuality is measured by the cost difference $J(c'.\,x) - J(A(c).\,x)$: the smaller this value, the "closer" the counterfactual scenario is to what actually occurred. This replaces Lewis's similarity metric on possible worlds with a cost-geometric measure grounded in the functional J.

11.4 Falsifiability

The RS modal grounding is empirically constrained in three ways:

1. **Uniqueness of necessity.** If a physical system were found to have multiple zero-defect states ($J(x) = 0$ for $x \neq 1$), the uniqueness theorem would be falsified. RS predicts that every physical observable, when expressed as a ratio to its natural unit, has $x = 1$ as its unique zero-cost value.

2. **Positivity constraint.** If a physical ratio x were measured to be non-positive, the possibility/impossibility boundary would be falsified. RS predicts all physical ratios are strictly positive.

3. **Cost monotonicity.** RS predicts that actualization decreases cost (Proposition 10.12). A physical process that systematically increased J-cost without external coupling would falsify the framework.

12 Conclusion

Modal logic is grounded in the cost functional $J(x) = \frac{1}{2}(x+x^{-1})-1$, the unique normalized solution to the Recognition Composition Law. The key results:

1. Necessity is the unique J-minimizer: only $x = 1$ is necessary (Theorem 4.1).

2. Possibility is positive ratio: every $x > 0$ is possible (Theorem 5.1).

3. Impossibility is the positivity boundary: $x \leq 0$ (Theorem 6.1).

4. Actuality is the J-minimum, not an indexical label (Theorem 7.1).

5. The implication chain necessary $\Rightarrow$ actual $\Rightarrow$ possible holds (Corollary 8.4).

6. Every possible configuration is accessible to the necessary one via cost-decreasing paths (Theorem 8.5).

7. Modal distance $d(x) = J(x)$ quantifies departure from actuality (Theorem 9.1).

8. Propositional operators $\Box$ and $\Diamond$ satisfy modal duality, axiom K, and weak axiom T (Propositions 10.8–10.10).

This gives modal metaphysics physical content without ontological inflation. "Necessary" means J-forced. "Possible" means finite cost. "Actual" means J-minimum. Modal distance is J-defect.

The framework entails a form of necessitarianism grounded in cost geometry: the actual world is the unique necessary one, existence is cheaper than non-existence, and the entire modal structure emerges from a single functional equation.

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